

Exercise 04, Computer Vision

Report :

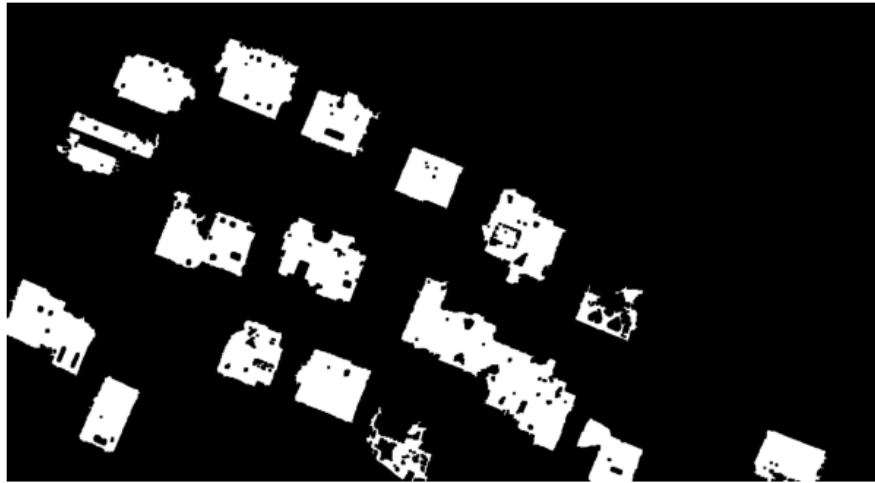
The solution starts with segmentation process, where vegetation is removed using the Normalized difference Vegetation index [NDVI] and roads are filtered out based on low HSV saturation levels, the rest of the pixels are then classified using K-Means clustering technique to identify bright building segments, which are also further cleaned via morphological opening and closing.

In the refinement step, the boundary accuracy is improved using Active Contour Model [Snake], which iteratively optimizes the building contours by minimizing the internal [for continuity and curvature] energy and external [image gradients] energy function.

For the vectorization process, the refined segment mask's contours are simplified and regularized to implement orthogonal [90 degree] angles, making sure the output consists of clean, rectilinear polygons, overlaid on the original image, representing the geometry of the buildings.

The solution achieves an intersection over Union [IoU] score of 79.45%.

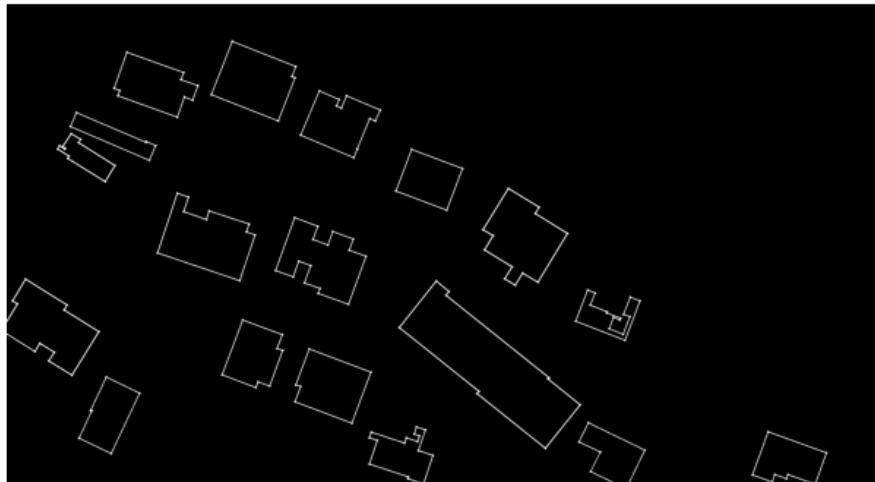
(a) Segmentation



(b) Refined Segmentation



(c) Vectorized Map



(d) Final Result

